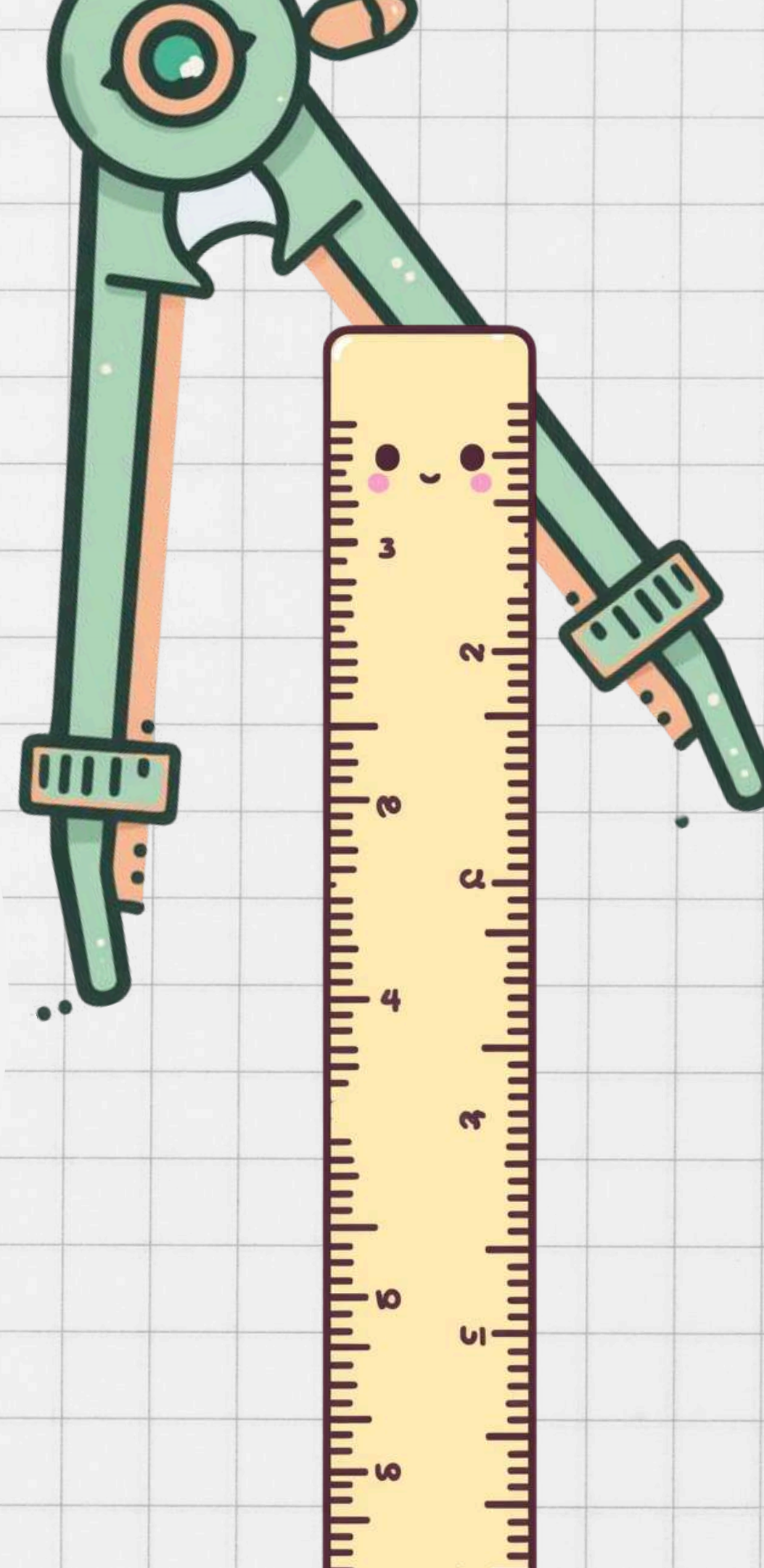


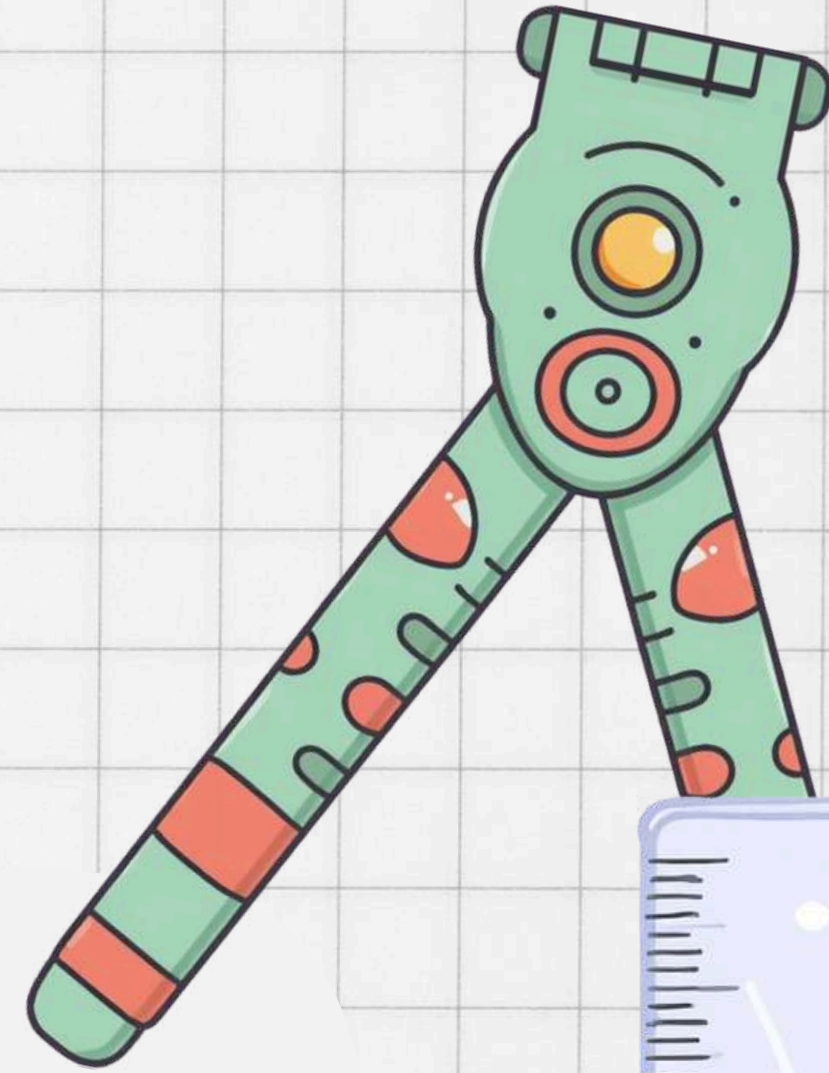
What Are Constructions?



Geometric constructions are drawings made using only a compass and a straight ruler — no measurements allowed! They help us create perfect shapes, lines, and angles. Constructions teach us precision and logical thinking, skills we use in art, architecture, and everyday problem-solving.



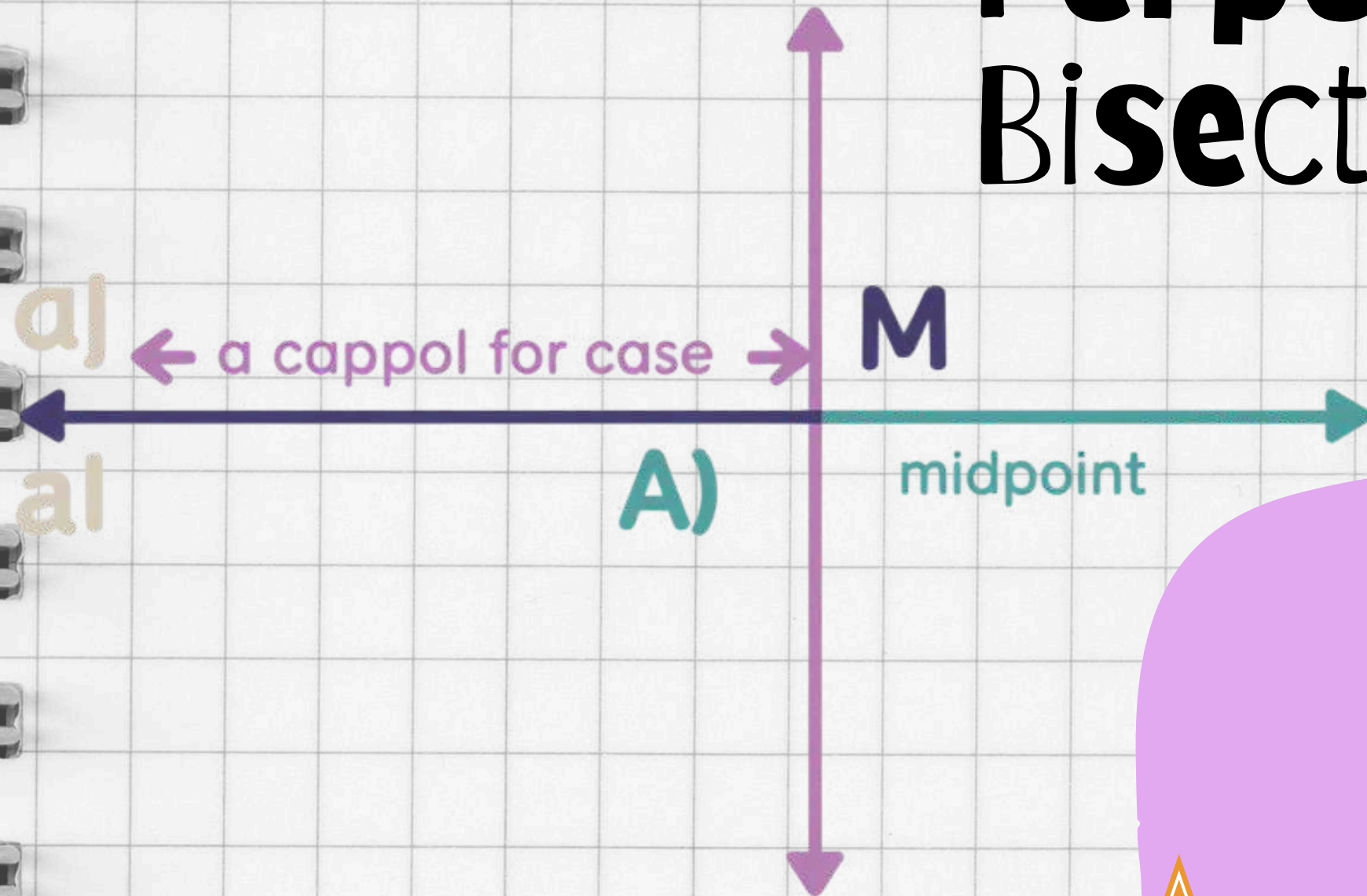
Tools for Constructions



Every great construction needs the right tools! A compass draws perfect circles and arcs. A ruler helps draw straight lines and measure lengths. A sharp pencil makes precise marks. An eraser fixes any mistakes along the way.



Perpendicular Bisector: What Is It?

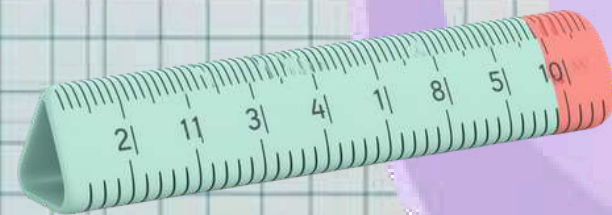
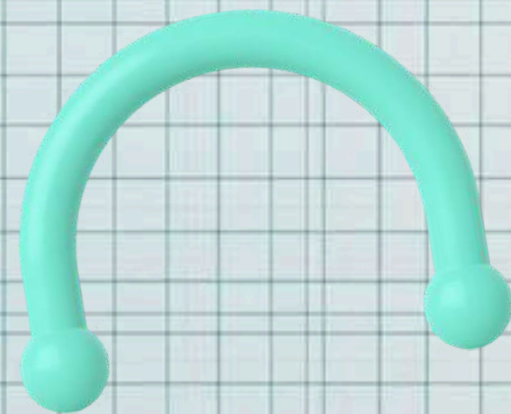


A perpendicular bisector is a special line that cuts a line segment into two equal parts at exactly 90 degrees. It passes through the midpoint and stands perfectly upright!



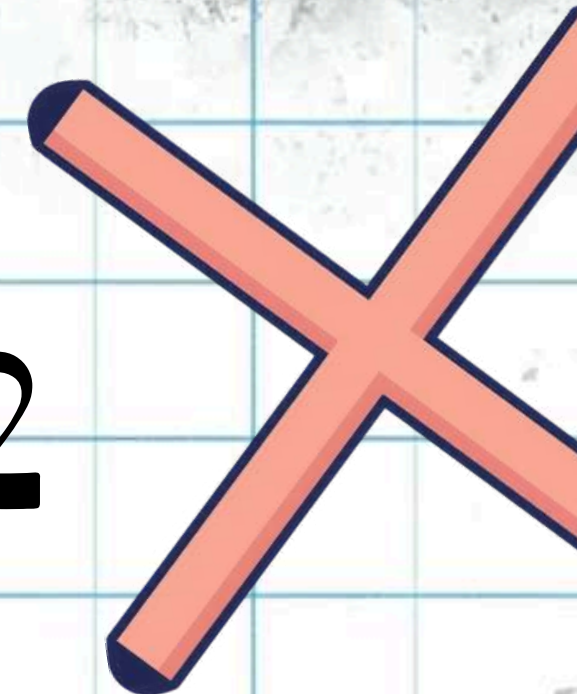
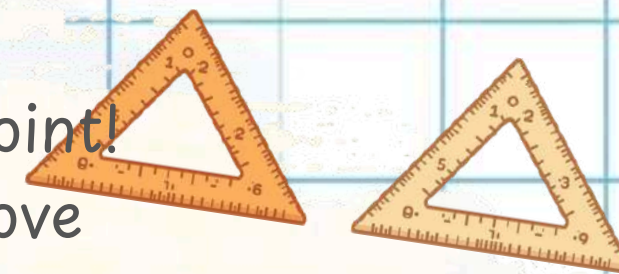
Perpendicular Bisector: Step 1

Place your compass point at one endpoint of the line segment. Open the compass wider than half the segment length. Draw two arcs — one above and one below the line segment. Keep the compass width the same!



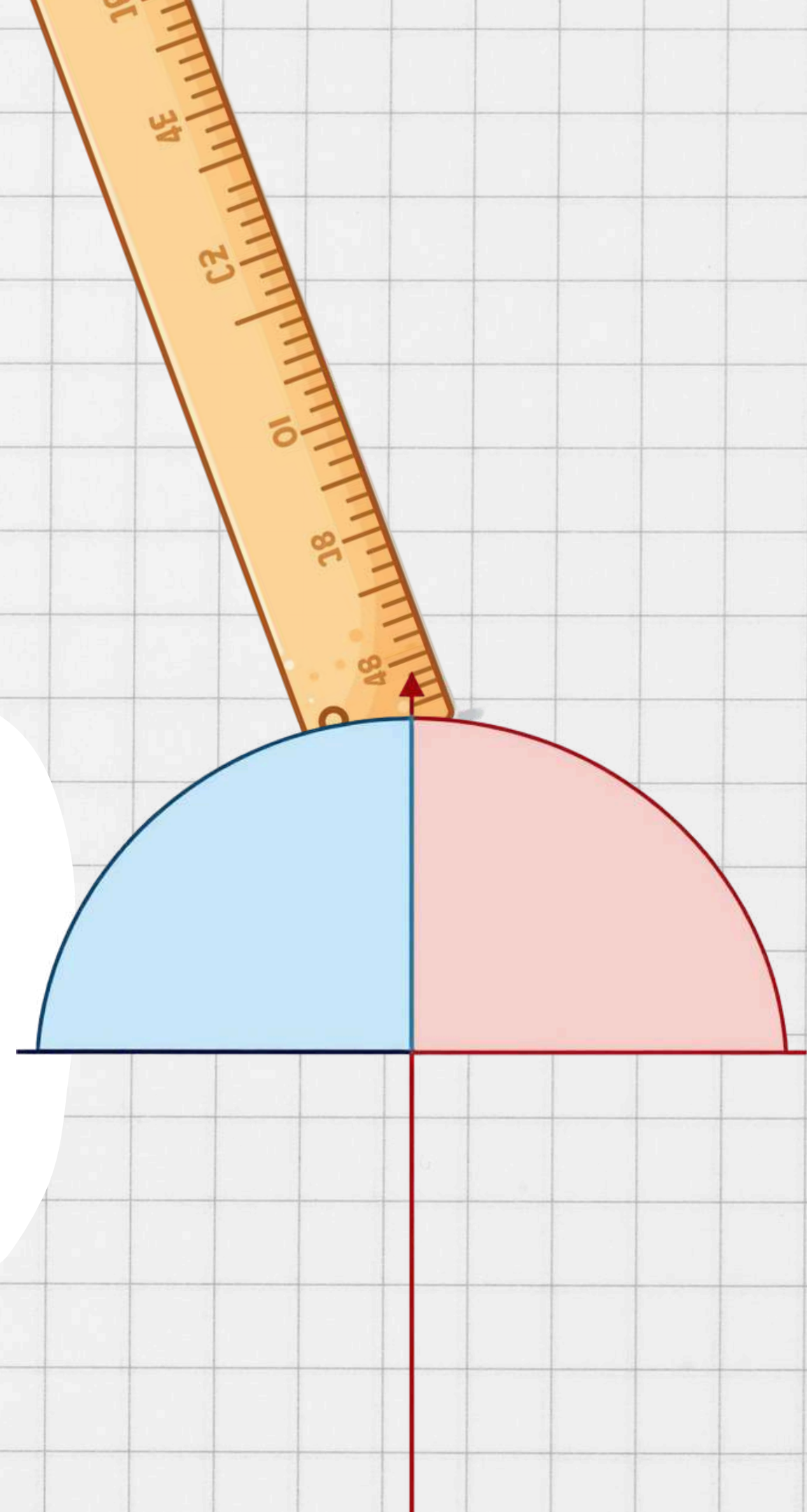
Perpendicular Bisector: Step 2

Now repeat the same process from the other endpoint!
Place your compass at point B and draw arcs above and below the line segment. Make sure the arcs cross the first set of arcs to create two intersection points.



Step 3: Draw the Bisector

Draw a straight line through the two intersection points where your arcs crossed. This line is your perpendicular bisector! It cuts the original line segment exactly in half at a 90° angle.

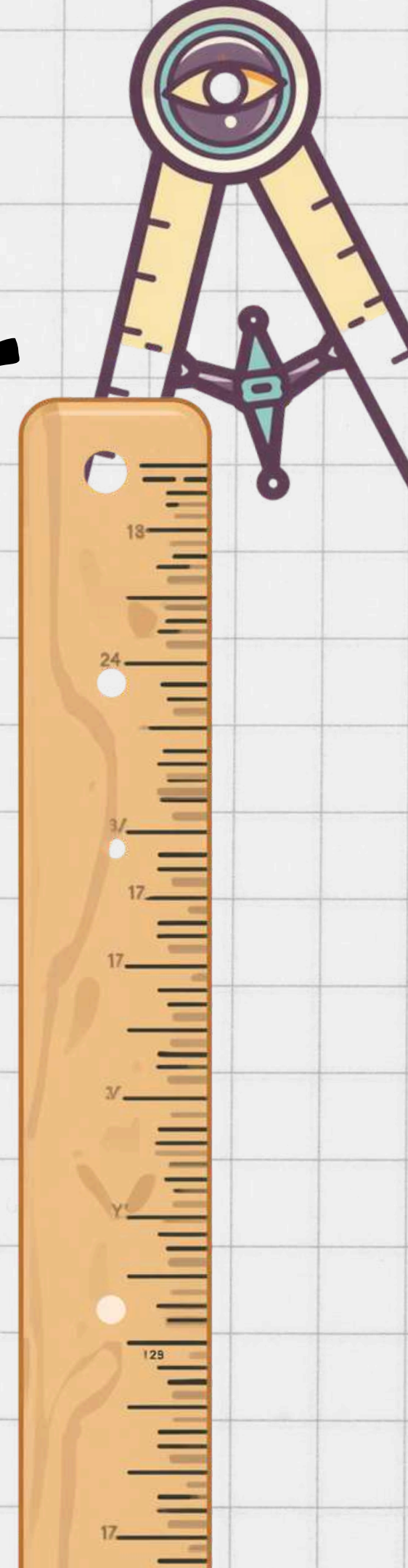


Practice: Perpendicular Bisector

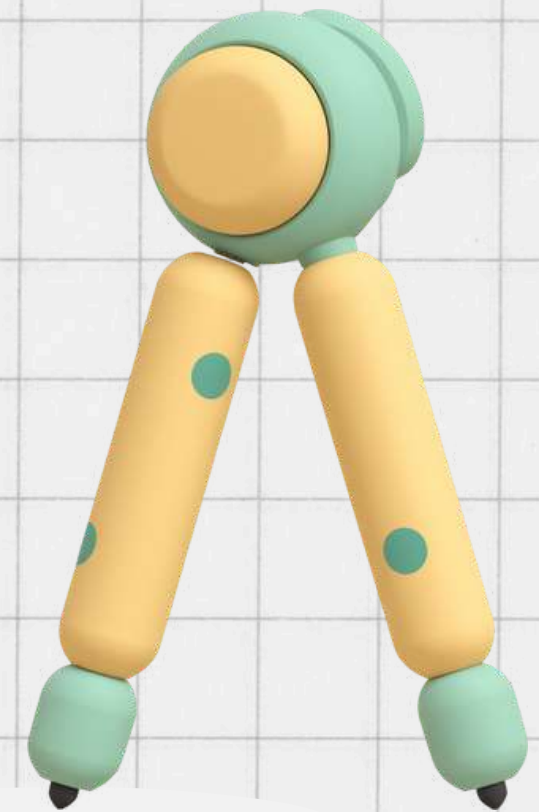
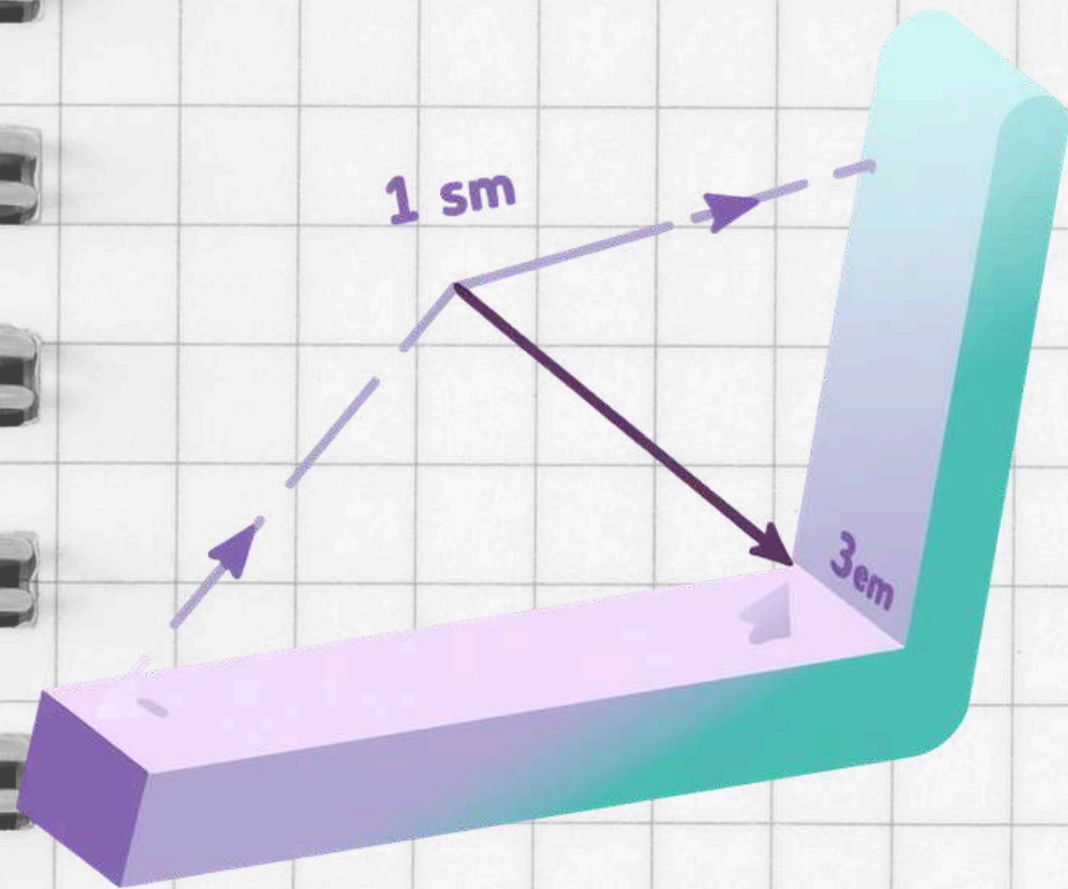
B

Try these exercises! For each line segment below, construct the perpendicular bisector using your compass and ruler:

1. Line segment $AB = 8\text{ cm}$
2. Line segment $PQ = 6\text{ cm}$
3. Line segment $XY = 10\text{ cm}$



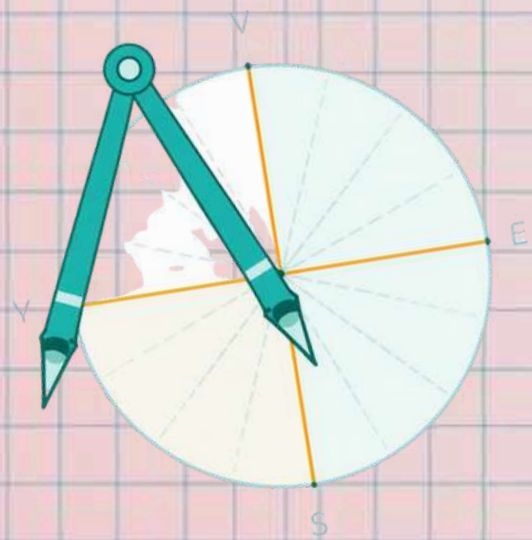
Angle Bisector: What Is It?



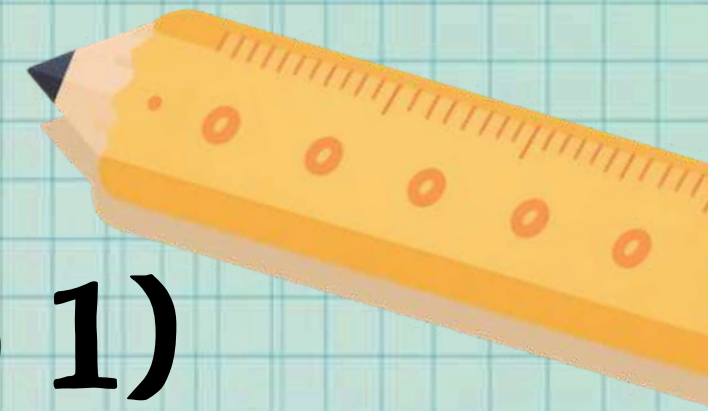
An angle bisector is a ray that divides an angle into two equal parts. It starts from the vertex and passes through the angle, creating two smaller angles of exactly the same measure.



How to Construct an Angle Bisector (Step 1)



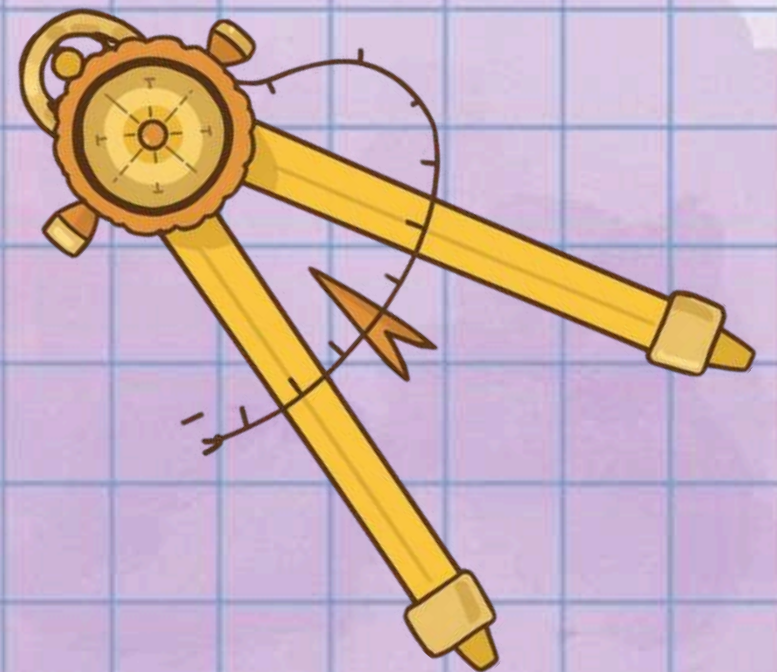
Step 1: Place the compass point at the vertex of the angle. Draw an arc that crosses both rays of the angle. These intersection points will be used in the next step.



Angle Bisector: Step 2

2

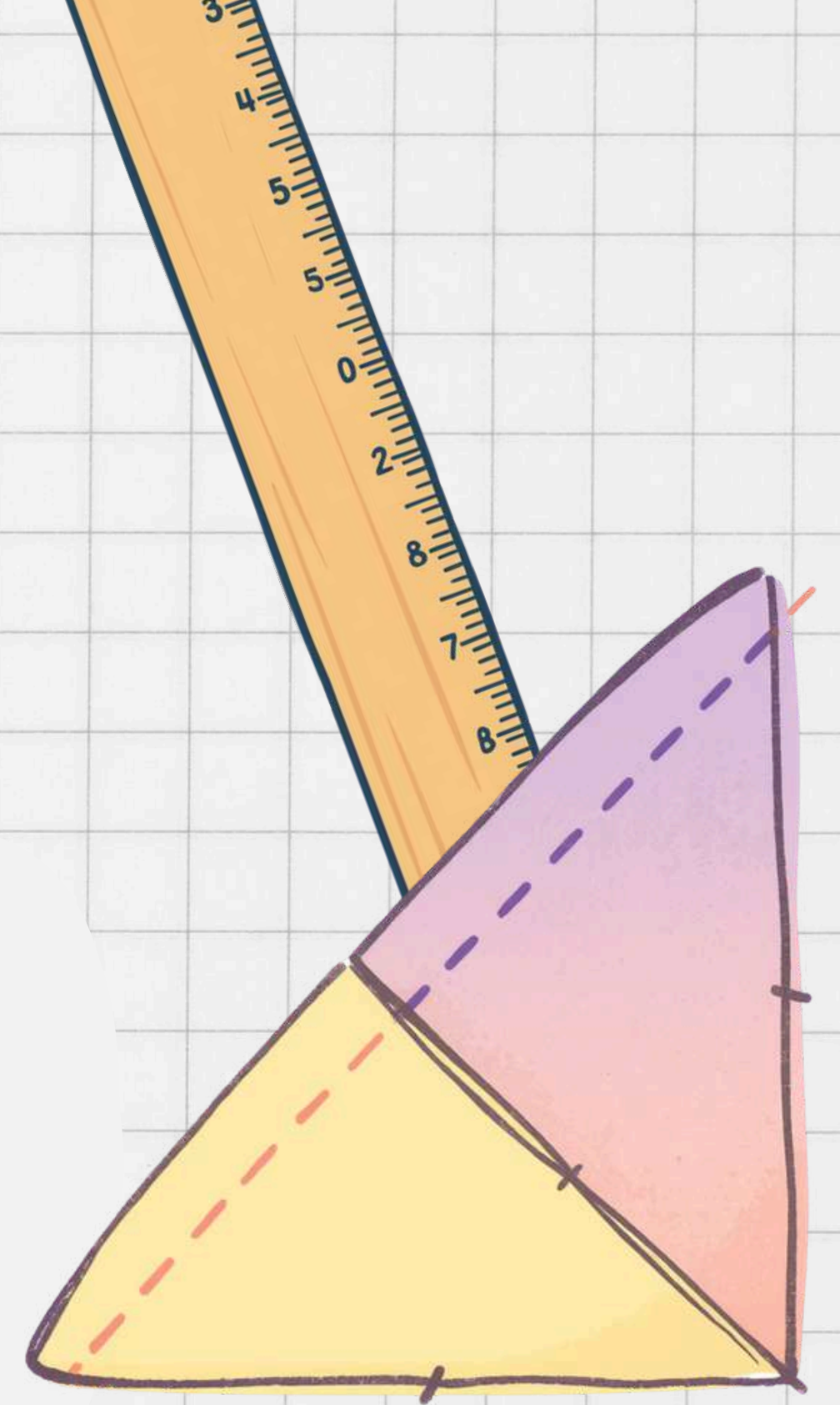
From each intersection point on the rays, place your compass and draw arcs inside the angle. Make sure these two arcs cross each other — this intersection point is very important for the next step!



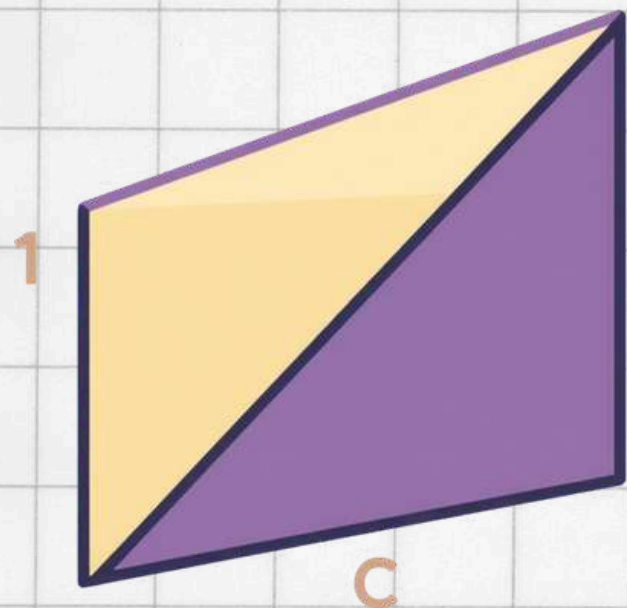
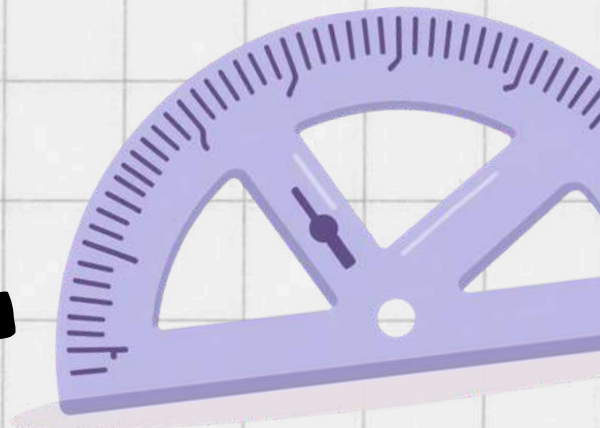
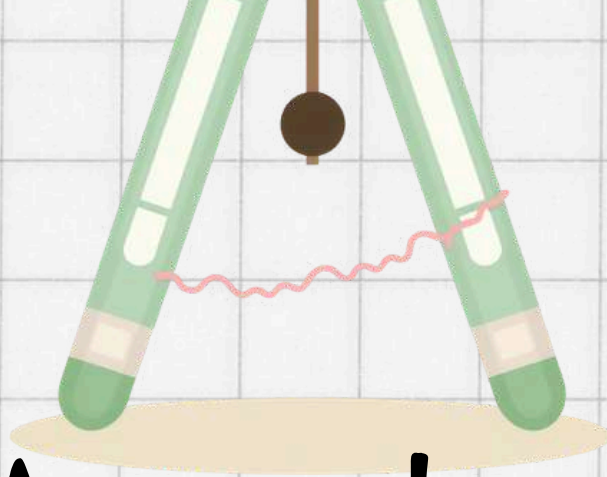
How to Construct an Angle Bisector (Step 3)



Step 3: Draw a straight ray from the vertex of the angle through the intersection point where your two arcs meet. This ray is your angle bisector — it divides the original angle into two perfectly equal parts!

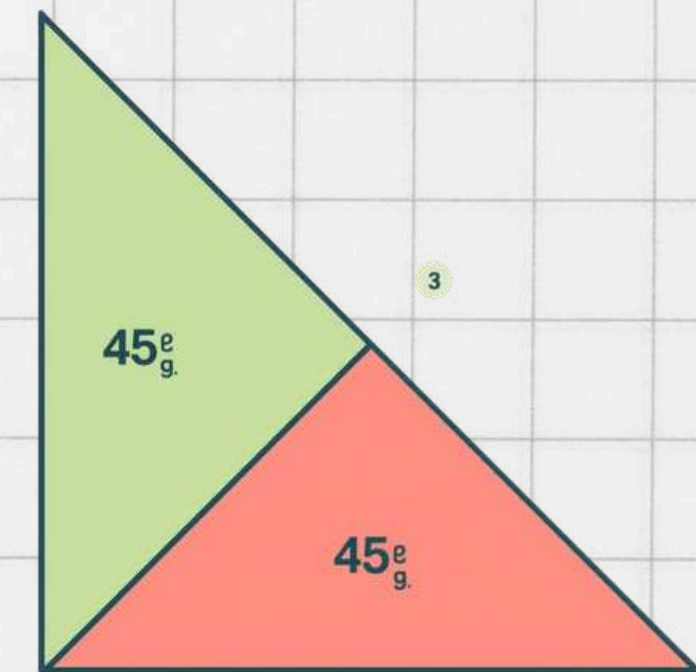


Practice: Angle Bisector

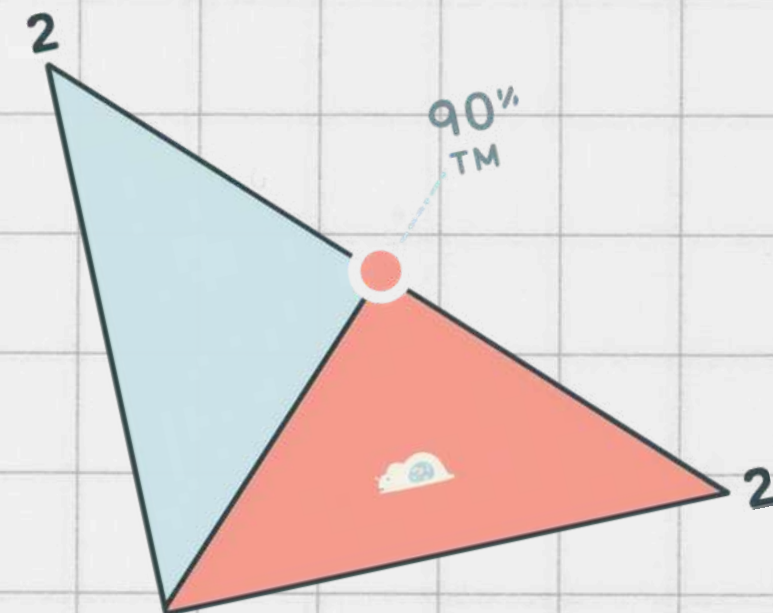


60

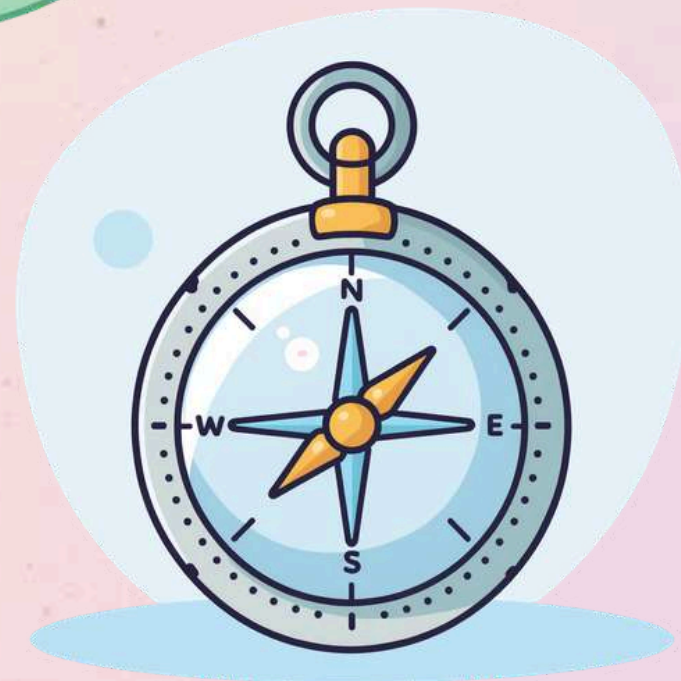
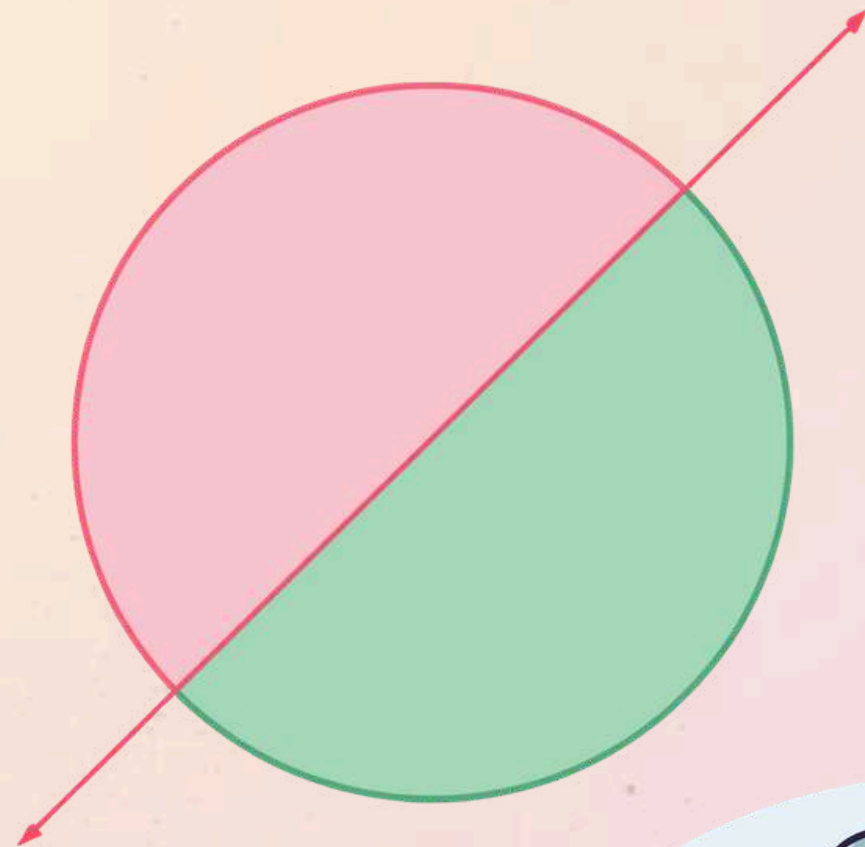
Try constructing angle bisectors for these three angles! Use your compass and ruler to divide each angle into two equal parts. Remember: arc from vertex, arcs from intersections, then draw the bisector ray.



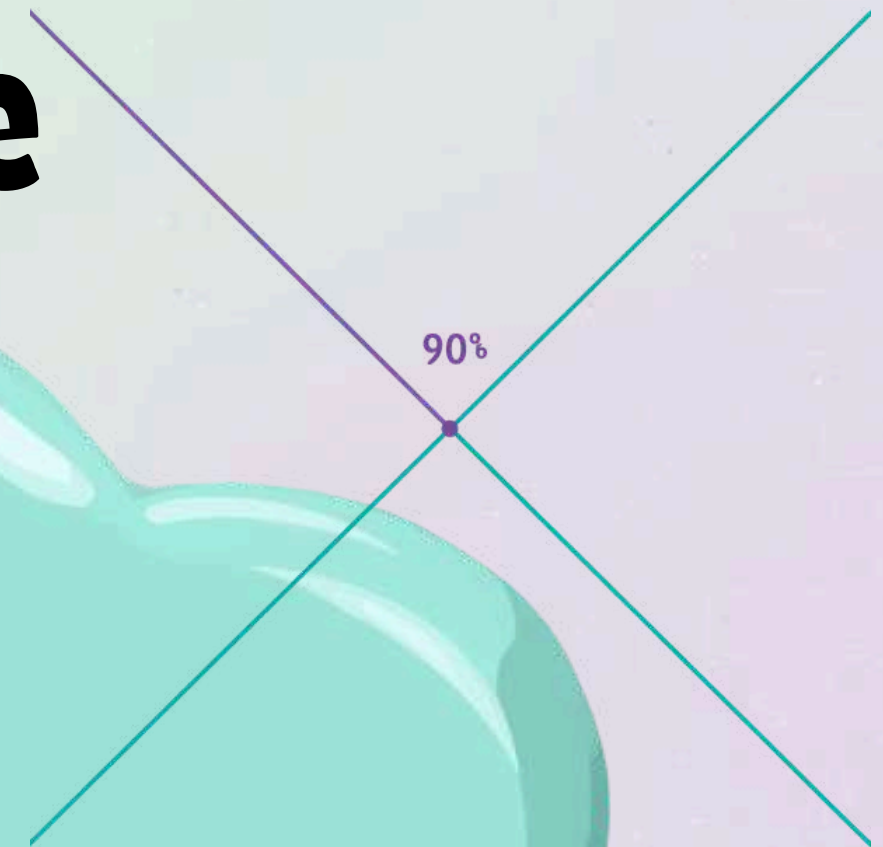
2



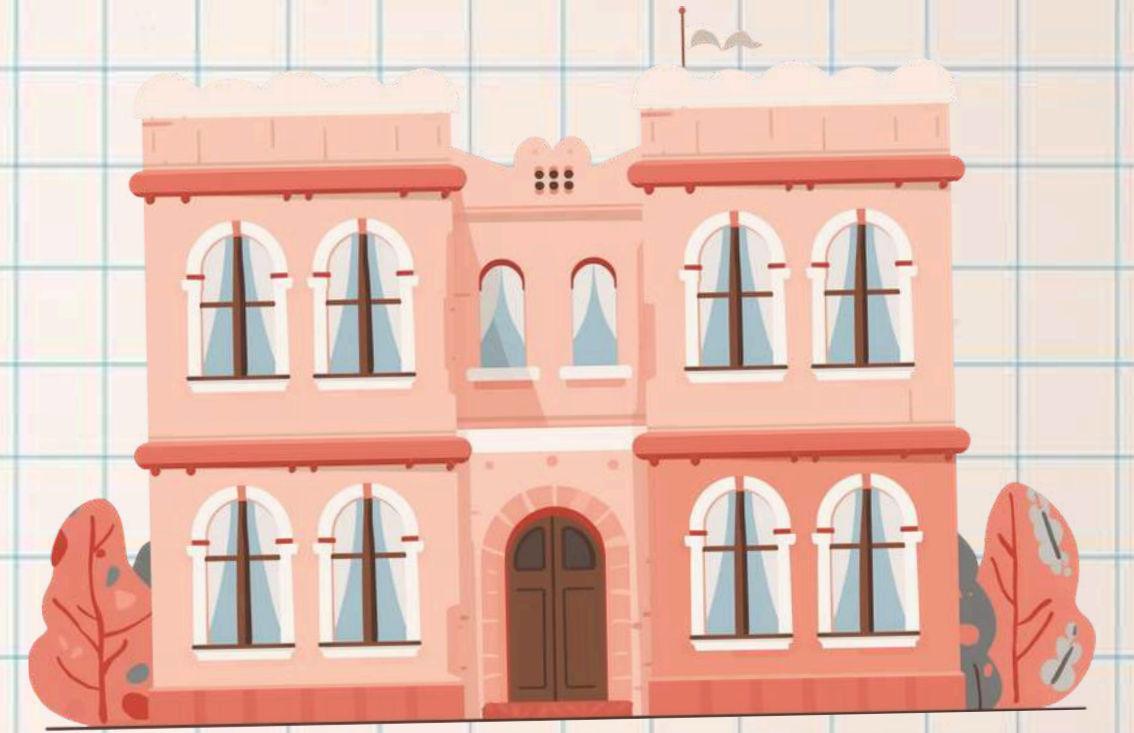
Perpendicular Bisector vs Angle Bisector



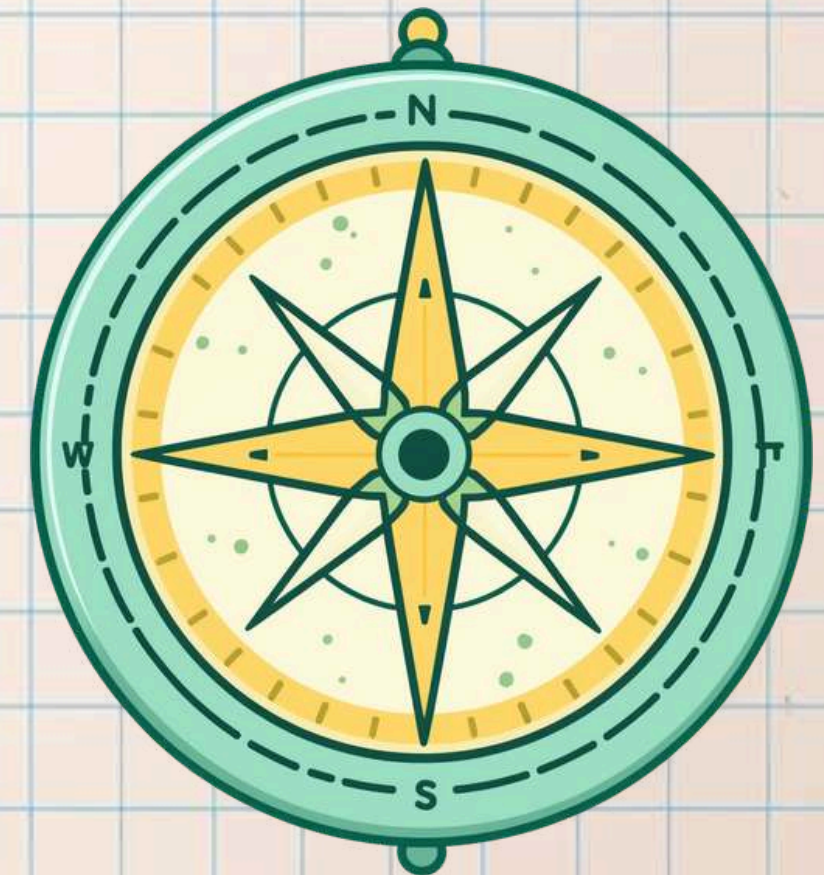
Let's compare! A perpendicular bisector divides a line segment into two equal parts at 90° . An angle bisector divides an angle into two equal parts. Both constructions use the same tools—compass and ruler!



Real-Life Uses of Bisectors



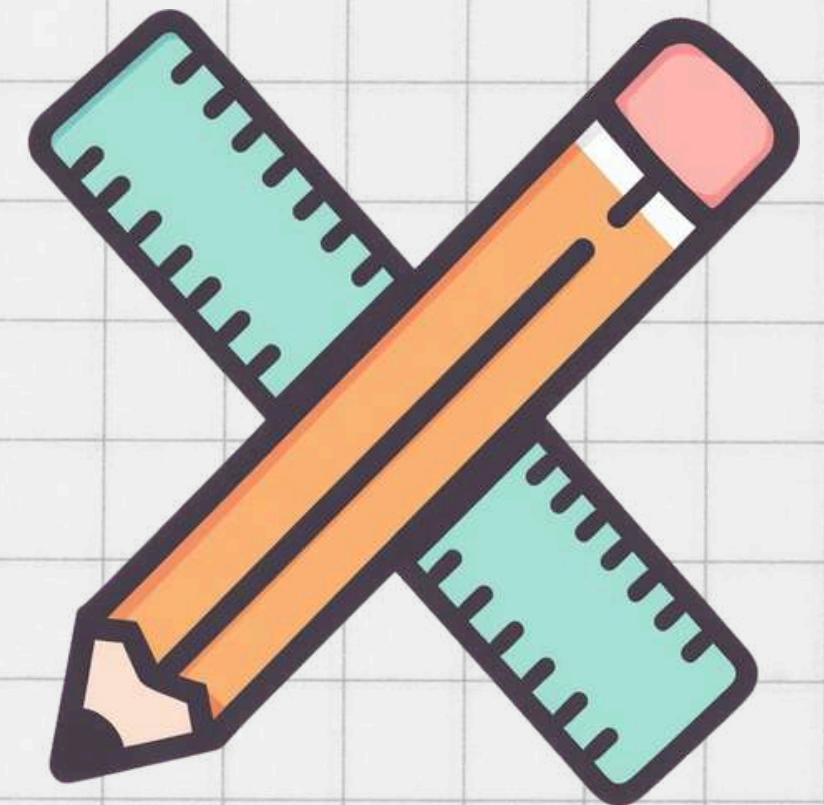
Bisectors are everywhere! Architects use them to design balanced buildings. Engineers apply them to build bridges and roads. Artists create symmetrical patterns and designs. Navigators use bisector principles to find directions and plot courses on maps.



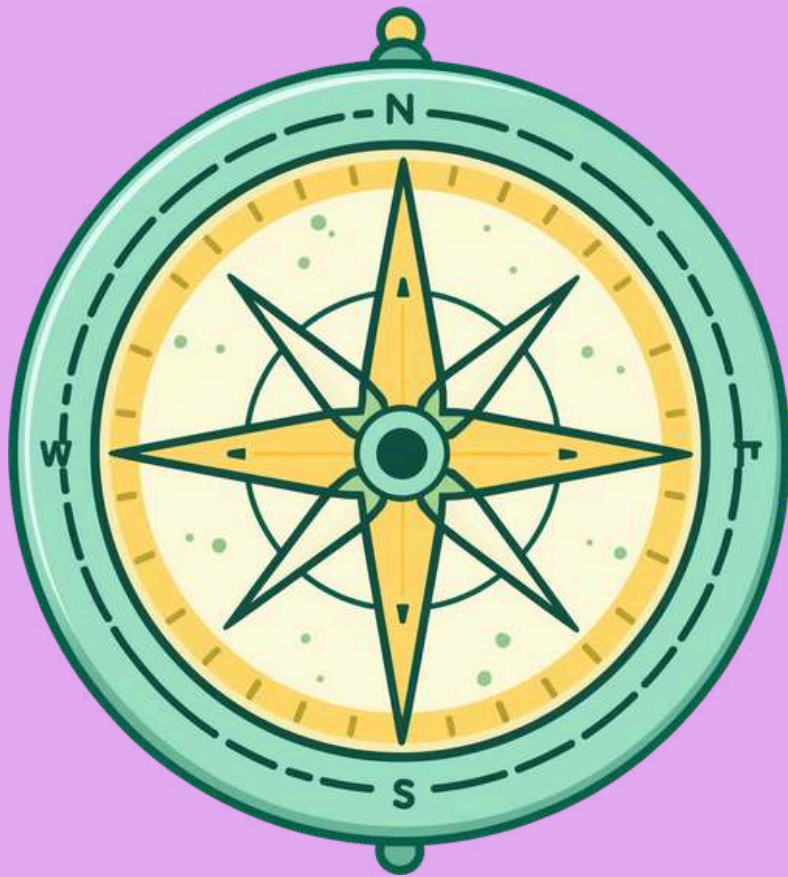
Summary & What's Next?



- Perpendicular bisector: A line dividing a segment into two equal parts at 90°
- Angle bisector: A ray dividing an angle into two equal parts
 - Both constructions use compass and ruler
 - Practice more problems in your workbook!



Inspiring the Future



Mathematics is more than a subject—it is a way of thinking that shapes the future. With creativity, logic, and imagination, these icons remind us that numbers will always guide us toward discovery.